

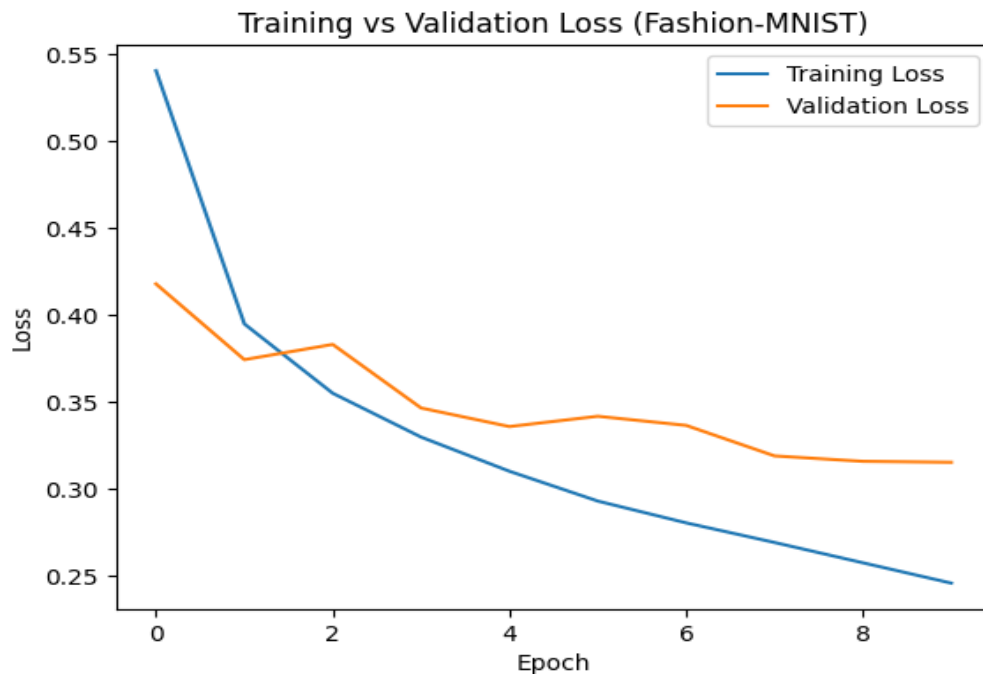
Assignment 1: Deep Learning – Performance Analysis on Fashion-MNIST

Dataset and Experimental Setup

All experiments were conducted on the Fashion-MNIST dataset, consisting of 28×28 grayscale images belonging to 10 clothing categories. The original training set of 60,000 samples was split into 80% training and 20% validation subsets. The standard test set of 10,000 samples was used only for final evaluation. A fully connected neural network was trained using cross-entropy loss and the Adam optimizer. Training and validation loss and accuracy were recorded at the end of each epoch.

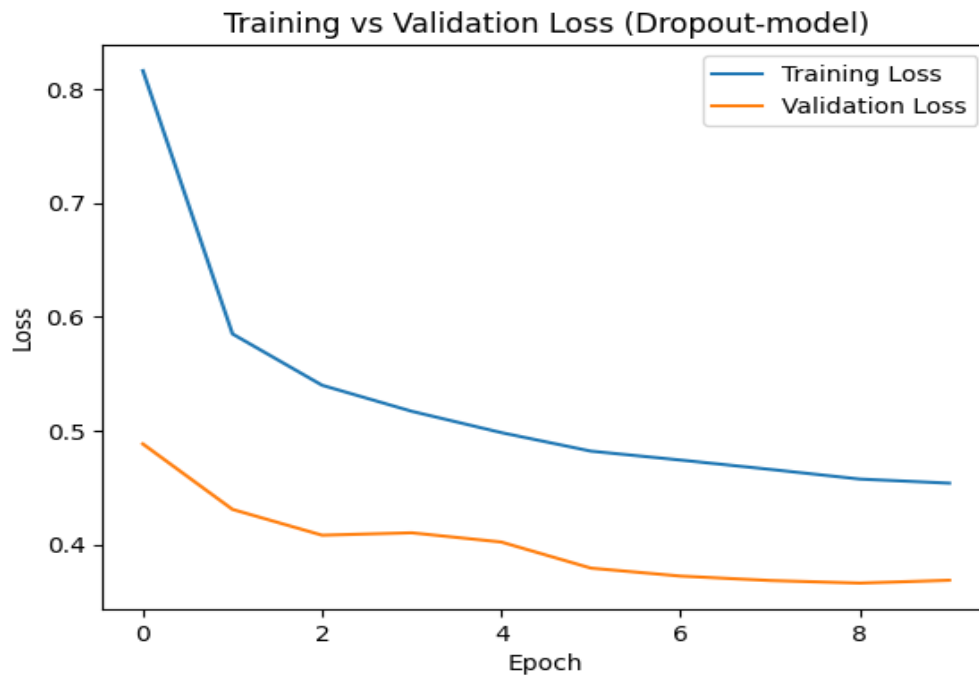
Question 1: Learning Behaviour of the Baseline Model

Figure 1 shows the training and validation loss for the baseline model. The training loss decreases steadily across epochs, indicating that the model is effectively learning patterns from the training data. The validation loss also decreases initially, but begins to plateau after a few epochs. This behaviour suggests the onset of mild overfitting, where the model continues to improve on the training set while generalisation improvements slow down. Training accuracy remains consistently higher than validation accuracy, which is expected in supervised learning. The final test accuracy closely matches the validation accuracy, indicating that the validation set provides a reliable estimate of the model's generalisation performance.



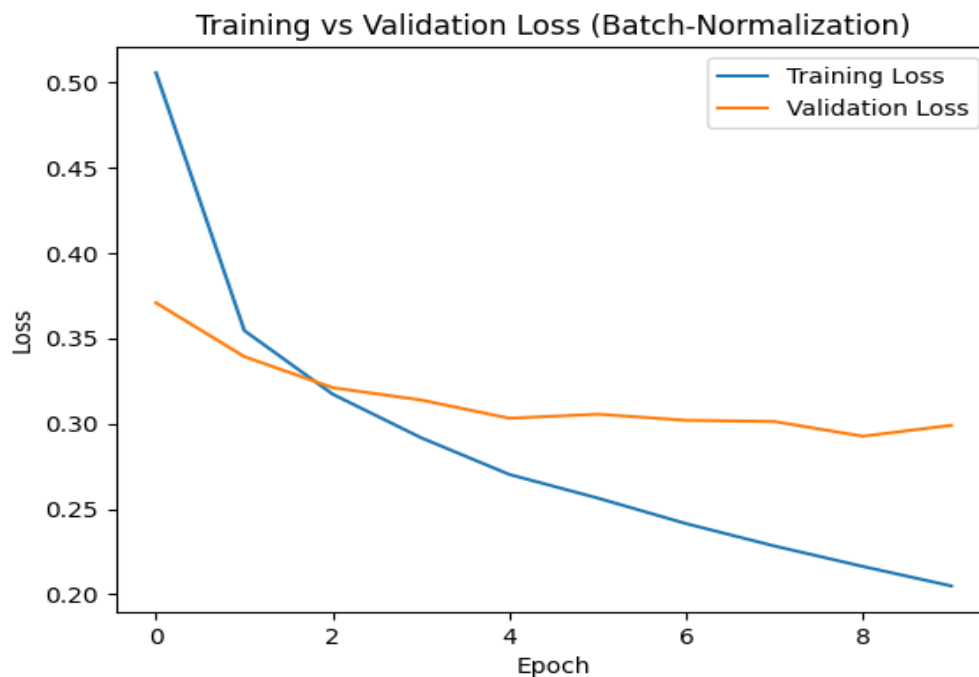
Question 2: Effect of Dropout on Overfitting

To address overfitting observed in the baseline model, Dropout layers were introduced after the hidden layers. As shown in Figure 2, the training loss for the Dropout model is higher compared to the baseline model, indicating that the network finds the learning task more difficult due to random neuron deactivation. However, the validation loss is lower and more stable. Compared to the baseline model, the gap between training and validation loss is reduced, demonstrating that Dropout effectively limits overfitting. By preventing the network from relying on specific neurons, Dropout encourages the learning of more robust and generalisable features.



Question 3: Effect of Batch Normalisation

Batch Normalisation layers were incorporated to stabilise and accelerate training. Figure 3 illustrates that the model with Batch Normalisation converges faster than the baseline model, as evidenced by the more rapid decrease in training loss during early epochs. The validation loss curve is smoother and shows less fluctuation, indicating improved training stability. Compared to the model without Batch Normalisation, validation performance improves earlier and remains consistent. This demonstrates that Batch Normalisation mitigates internal covariate shift, enabling more efficient optimisation and improved generalisation.



Conclusion

The baseline model demonstrates effective learning but exhibits mild overfitting. Introducing Dropout reduces overfitting by narrowing the gap between training and validation performance, while Batch Normalisation improves convergence speed and training stability. Together, these techniques significantly enhance the robustness and generalisation capability of the neural network on the Fashion-MNIST dataset.